

WORKING PAPER

Reserves Expansion and Self-fulfilling Debt Crisis

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Abstract

I study how central bank balance sheet policies influence the credit spread of sovereign debt. In my model, multiple debt price equilibria arise due to investors' self-fulfilling beliefs about the future government debt path. Quantitative easing (QE), by increasing the supply of reserves — which are more senior liabilities — in exchange for long-term debt, dilutes existing bondholders and can trigger a shift to the default equilibrium if implemented as a one-time unexpected policy. In contrast, a credible commitment to expand the balance sheet in response to rising debt can anchor expectations and eliminate the default equilibrium by moderating the perceived future debt trajectory.

JEL Codes: E58, G12, H63.

Keywords: Quantitative easing, sovereign debt crises, self-fulfilling equilibria, reserves seniority, balance sheet policy.

1 Introduction

Sovereign bonds sometimes suffer sudden price drops that cannot be justified by fundamentals that move at slower rates. The literature on self-fulfilling debt crises [Calvo, 1988, Cole and Kehoe, 2000, Aguiar et al., 2022, Lorenzoni and Werning, 2019] views the phenomenon as a sudden equilibrium shift: if investors believe the government has a high probability of default, bond prices fall; facing higher borrowing costs, the government finds it harder to roll over debt and is more likely to default, which in turn validates the initial investor belief. To eliminate the undesirable crisis equilibrium, the government could take fiscal actions such as cutting current debt, committing to stronger fiscal tightening rules, or lengthening debt duration.

In this paper I study a different stabilizing instrument: *balance sheet policy*. I argue that a credible commitment to expand central bank reserves — in exchange for long-term bonds — can anchor expectations and eliminate the default equilibrium by moderating the perceived future debt trajectory, even though a one-time unexpected reserves expansion may *worsen* the situation. Although QE is typically viewed as a monetary instrument, a reserves-for-bonds exchange effectively substitutes short-term real liabilities for longer-term liabilities. Crucially, reserves differ from short-term government bonds in two ways: (i) investors do not doubt the value of reserves and treat them as senior debt that the government always services first when a fiscal crisis occurs; (ii) issuing reserves to buy back bonds slows the growth rate of long-term debt, buying time for fiscal adjustment and reducing perceived default risk.

The model is built on the continuous-time sovereign-debt framework of Lorenzoni and Werning [2019]. The government receives a flow of fiscal revenues and runs a state-contingent expenditure rule that links primary surpluses to the current debt level. It issues long-term bonds b (geometrically decaying consols) and reserves m . Debt resolution risk arrives at Poisson rate λ : at the resolution stage the government services as much debt as fiscal capacity Z allows, with reserves holding priority over bonds. The main results are three propositions.

Multiple equilibria (Proposition 1). Under the baseline parameterization there are two steady states. The left (stable) steady state supports a good equilibrium $Q^+(b, m)$ in which bond prices converge to a positive value; the right (unstable) steady state is the entry point

into a bad equilibrium $Q^-(b, m)$ in which prices fall to zero and debt explodes. The crisis region $[b^-, b^+]$ contains all initial debt levels for which both equilibria are self-fulfilling.

Adverse one-time QE (Proposition 2). An unexpected large-scale bond purchase financed by *senior* reserves reduces $q(m, b)$: issuing senior liabilities dilutes existing bond holders, lowering the expected recovery value $\Psi(b, m)$. Beyond a threshold reserves injection the economy can be pushed from the good to the bad equilibrium even if the initial debt level was outside the crisis region.

Benign expansion promises (Proposition 3). A credible commitment to expand reserves at rate $\beta > 0$ in response to fiscal need raises *both* Q^+ and Q^- and shifts the crisis region $[b^-, b^+]$ rightward, regardless of reserves seniority. The mechanism is purely through debt-growth moderation: faster reserves issuance crowds out bond issuance, slowing debt accumulation and reducing the probability of hitting the crisis region.

The paper proceeds as follows. Section 2 sets out the model. Section 3 characterizes the equilibrium pricing functions and presents the three propositions. Section 4 concludes.

Related literature. The paper is most closely related to [Lorenzoni and Werning \[2019\]](#), whose continuous-time framework I adopt. [Calvo \[1988\]](#) and [Cole and Kehoe \[2000\]](#) are the classical self-fulfilling debt crisis models; [Aguiar et al. \[2022\]](#) revisit the mechanism in a quantitative setting. On quantitative easing, [Krishnamurthy and Vissing-Jorgensen \[2011\]](#) study interest-rate channels; [Haddad et al. \[2024\]](#) and [Haddad et al. \[2025\]](#) examine asset-purchase rules and conditional policy promises. The safe-haven status of US Treasuries is studied in [He and Krishnamurthy \[2020\]](#) and [He et al. \[2022\]](#); [Jiang et al. \[2024\]](#) document the US public debt valuation puzzle that motivates the fiscal channel explored here. [Du et al. \[2024\]](#) surveys what quantitative tightening has taught us about the QE transmission mechanism.

2 Model

The setup follows [Lorenzoni and Werning \[2019\]](#). The economy is continuous-time, real, and features a consolidated government and a representative investor.

2.1 The Government

Fiscal rule. The government receives a constant flow of fiscal revenue T_t and runs a state-contingent expenditure rule e_t . Net fiscal surplus is

$$z_t = T_t - e_t = h(b_t) = \min\{\alpha_0 + \alpha_1 b_t, \bar{z}\}, \quad (1)$$

where $\alpha_0, \alpha_1 > 0$ so that the surplus increases with the debt level, and $\bar{z}$ is the maximum fiscal capacity. This reduced-form rule captures the idea that fiscal policy tightens when debt is high and can be micro-founded from the government's optimization problem.

Liabilities. The government issues two types of liabilities: long-term bonds b_t and reserves m_t . Time is continuous; bonds decay geometrically at rate δ , paying a dividend flow κb_t where $\kappa = \delta + \rho$ normalizes the default-free bond price to one. Reserves pay a fixed real interest rate r , endogenously determined in equilibrium. Bond price $q_t = q(m_t, b_t)$ is a function of the two state variables.

Budget constraint. The government's flow budget constraint is

$$z_t + q(m_t, b_t)(\mu_t^b + \delta b_t) + \mu_t^m = \kappa b_t + m_t r, \quad (2)$$

where μ_t^b and μ_t^m are the rates of bond and reserves issuance. The left-hand side is resources available (surplus plus proceeds from new issuance); the right-hand side is debt service obligations (coupons plus reserves interest).

Reserves expansion rule. The government commits to a reserves expansion rule

$$\mu_t^m = \beta_t q_t \mu_t^b \mathbf{1}_{\{m_t < \bar{m}\}}, \quad \beta_t \geq 0, \quad (3)$$

so that a fraction $\beta/(1 + \beta)$ of financing needs is met by issuing reserves (senior short-term liabilities) rather than bonds. The parameter β_t is constant in the baseline; its role as a policy instrument is analyzed in Section 3.3. The government may also conduct a one-time unexpected large-scale purchase $\Delta b_t = b_t - b_{t-}$ financed by $\Delta m_t = m_t - m_{t-}$ reserves such that

$$q(m_t + \Delta m_t, b_t + \Delta b_t) \Delta b_t + \Delta m_t = 0. \quad (4)$$

The money and debt dynamics in continuous time are

$$db_t = \mu_t^b dt + \Delta b_t, \quad dm_t = \mu_t^m dt + \Delta m_t. \quad (5)$$

Debt-resolution stage. When $q_t = 0$ the government is forced to default and exit the debt market; it finances future fiscal needs by issuing reserves only. More fundamentally, default risk originates from a *debt-resolution stage* that arrives at Poisson rate λ . At the resolution stage, the government can no longer issue new debt; existing bonds stop decaying but grow in face value at the short rate. From entry time s onward, bond face value evolves as

$$db_\tau = b_\tau r_\tau d\tau - (z_\tau - \theta m_\tau r_\tau) d\tau, \quad \tau > s, \quad (6)$$

where θ is the *reserves seniority* parameter ($\theta = 1$ means the government must service reserves first; $\theta = 0$ means reserves have no priority). The government repays all debt using future fiscal surpluses $\{z_\tau\}_{\tau>s}$, but must first cover θ of outstanding reserves interest.

Fiscal capacity and default. Let $Z_t = \int_t^{+\infty} (z_s - \theta m_t r_s) \exp(-\int_t^s r_u du) ds$ be the discounted sum of future net fiscal surplus net of reserves obligations. Z is log-normally distributed: $Z \sim \exp(\mathcal{N}(\mu_z, \sigma_z^2))$. If $Z_t - \theta m_t < b_t$, the government defaults on bonds and investors recover fraction ϕ of the shortfall. The expected bond value at the resolution stage is

$$\Psi(b, m) = 1 - F(b + \theta m) + \frac{\phi}{b} \int_{\theta m}^{b + \theta m} (Z - \theta m) dF(Z), \quad (7)$$

where F is the CDF of Z . The demand-side (investor indifference) condition reduces to the steady-state valuation

$$\Psi_b(b, m) < 0, \quad \Psi_m(b, m) < 0, \quad (8)$$

where both partial derivatives are negative: higher debt or higher senior reserves reduce the expected recovery value.

2.2 Investors

Investors are risk-neutral with discount rate ρ and solve the consumption-maximization problem

$$\mathbb{E}_t \left[\int_t^\infty c_u e^{-\rho(u-t)} du \right], \quad (9)$$

subject to their budget constraint. Investors hold bonds b_t^h and reserves m_t^h ; their positions evolve symmetrically to the government's issuance decisions. At the debt-resolution stage the investor consumes the bond flow payment, equal to $(z_t - \theta m_{t_0}^h r_t) \mathbf{1}_{\{b_{t_0}^h > 0\}} + m_t^h r_t$ without default, or $\phi(z_t - \theta m_{t_0}^h r_t) + m_t^h r_t$ with default.

Bounded rationality. Investors do not anticipate potential changes in monetary policy or balance sheet policy. They expect neither one-time large-scale purchases nor changes in the reserves-expansion speed β . When such a change is announced, investors update their expectations accordingly. This assumption isolates the *announcement effect* of a reserves expansion commitment from ongoing expectations of future balance sheet operations.

3 Equilibrium

An **equilibrium** is a continuum of $\{c_t, b_t^h, m_t^h, b_t, m_t, \Delta b_t, \Delta m_t, \Delta \beta_t, q_t, r_t, \beta_t, z_t\}_{t=0}^{\infty}$ such that: (i) the budget constraints of the government and investors hold; (ii) given announced government policies $\{\Delta b_s, \Delta m_s, \Delta \beta_s\}_{s \geq t}$, current asset prices (q_t, r_t) and their implied dynamics, the path $\{c_t, b_t^h, m_t^h\}$ solves investors' maximization; (iii) government liabilities follow the announced dynamics; and (iv) markets clear: $b_t^h = b_t$, $m_t^h = m_t$, $c_t + e_t = y_t$ (endowment).

In equilibrium investors are indifferent between holding reserves (at rate $r = \rho$) and bonds, so the bond's holding return must equal ρ :

$$r^b q = \kappa - \delta q + \lambda(\Psi(b, m) - q) + \dot{q}, \quad r^b = r = \rho. \quad (10)$$

Rearranging, the bond pricing ODE is

$$\dot{q} = (\rho + \delta + \lambda) q - \kappa - \lambda \Psi(b, m). \quad (11)$$

The debt dynamics (for $\beta = 0$, constant reserves) follow from the budget constraint (2):

$$\dot{b} = \frac{\kappa b + mr - z(b)}{q} - \delta b. \quad (12)$$

Equations (11)–(12) form a two-dimensional ODE in (b, q) . We impose a no-bubble condition: eligible equilibrium price paths either converge to the stable steady state ($q \rightarrow q_{ss} > 0$, $b \rightarrow b_{ss}$) or to $q = 0$ in finite time with debt exploding.

3.1 Multiple Equilibria

The steady-state conditions are $\dot{q} = 0$ and $\dot{b} = 0$:

$$\dot{q} = 0: \quad q = \frac{\kappa + \lambda \Psi(b, m)}{\rho + \delta + \lambda} \quad (\text{demand curve}), \quad (13)$$

$$\dot{b} = 0: \quad q = \frac{\kappa b + mr - z(b)}{\delta b} \quad (\text{supply curve}). \quad (14)$$

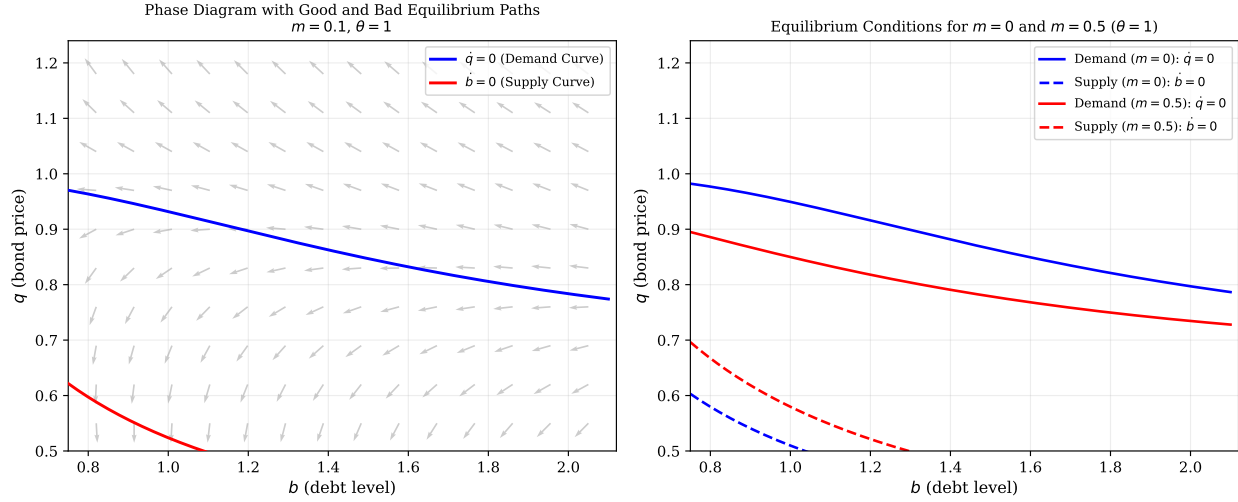


Figure 1: Phase Diagram with Equilibrium Price Paths. *Left*: vector field, $\dot{q} = 0$ demand curve (blue) and $\dot{b} = 0$ supply curve (red), good equilibrium path Q^+ (green) and bad equilibrium path Q^- (purple), for $m = 0.1, \theta = 1$. Circle (square) marks the stable (unstable) steady state. *Right*: equilibrium loci for $m = 0$ (blue) and $m = 0.5$ (red); the unstable steady state shifts left as reserves increase.

The demand curve is decreasing in b (higher debt raises default risk, lowering recovery Ψ). The supply curve can be non-monotone when $h'(b) > \kappa - \delta q$: for low debt the surplus grows with debt, requiring a higher bond price to balance the budget; for high debt the surplus is capped at $\bar{z}$, so a higher bond price is needed to carry the larger coupon burden.

Proposition 1. *The steady-state demand function is downward sloping: $\Psi_b(b, m) < 0$ and $\Psi_m(b, m) < 0$. The steady-state supply function can be non-monotone if $h'(b) > \kappa - \delta q$. Under the baseline parameterization there exist two steady states. The left steady state (b^-, q^-) is stable; the right steady state (b^+, q^+) is unstable. For any initial debt $b \in [b^-, b^+]$ there are two equilibria: a good equilibrium $Q^+(b, m)$ converging to (b^-, q^-) and a bad equilibrium $Q^-(b, m)$ along which $q \rightarrow 0$ in finite time.*

Figure 1 illustrates the phase diagram. The left panel shows the vector field and the two equilibrium paths for $m = m_0 = 0.1$ and $\theta = 1$. The good path Q^+ (green) approaches the stable steady state from below; the bad path Q^- (purple) diverges toward default from the unstable steady state. The right panel compares the equilibrium loci across two reserves levels ($m = 0$ and $m = 0.5$): higher senior reserves shift the demand curve down (lower recovery value) and the supply curve left (higher fiscal burden from reserves interest), moving the unstable steady state to the left.

3.2 Unexpected Bonds Purchase

We now examine the effect of an unexpected one-time bond purchase. The government announces a buy-back $(\Delta m, \Delta b)$ satisfying constraint (4), increasing m and reducing b . Starting from the good equilibrium $q(m_0, b_0) = Q^+(m_0, b_0)$, the post-purchase state is $(m_0 + \Delta m, b_0 + \Delta b)$.

Proposition 2 (Adverse QE). *If reserves are senior ($\theta = 1$), then $q(m + \Delta m, b + \Delta b) < q(m, b)$ for any $\Delta m > 0$. Buying bonds with senior reserves dilutes existing long-term bond holders. If before the policy $q(m, b) = Q^+(m, b)$, there exists a threshold $\bar{\delta}$ such that $q(m + \Delta m, b + \Delta b) = Q^-(m + \Delta m, b + \Delta b)$ whenever $\Delta m > \bar{\delta}$, i.e., no bond price level can sustain the good equilibrium.*

The intuition is as follows. Senior reserves reduce the net fiscal capacity available to bond holders: only $Z_t - \theta m_t$ can be used to repay bonds. A reserves injection lowers $\Psi(b, m)$ even after b falls proportionately, because the reserves-senior dilution effect dominates the debt-reduction benefit. If the injection is large enough, the new state $(m + \Delta m, b + \Delta b)$ lies entirely below the bad equilibrium curve, pushing the economy to default.

Figure 2 shows the Q^+ and Q^- pricing curves for $m \in \{0, 0.10, 0.15\}$. With money seniority (left panel), increasing m shifts both curves down substantially: a buy-back that was price-neutral without seniority can destroy the good equilibrium with seniority. Without seniority (right panel), the buy-back reduces b one-for-one, leaving bond prices almost unchanged — consistent with Proposition 2.

3.3 Reserves Expansion Rules

We now examine the effect of a credible commitment to expand reserves at rate $\beta > 0$. With an active expansion rule, debt dynamics become

$$\dot{b} = \frac{\kappa b + mr - z(b)}{q(1 + \beta)} - \delta b, \quad (15)$$

so the effective speed of debt accumulation is slowed by the factor $(1 + \beta)^{-1}$. The supply curve $\dot{b} = 0$ shifts:

$$q_\beta = \frac{\kappa b + mr - z(b)}{\delta b(1 + \beta)}, \quad (16)$$

which is lower than q_0 for any given b . Crucially, the demand curve $\dot{q} = 0$ is unchanged because Ψ depends on the *stock* of reserves and bonds, not on their growth rates.

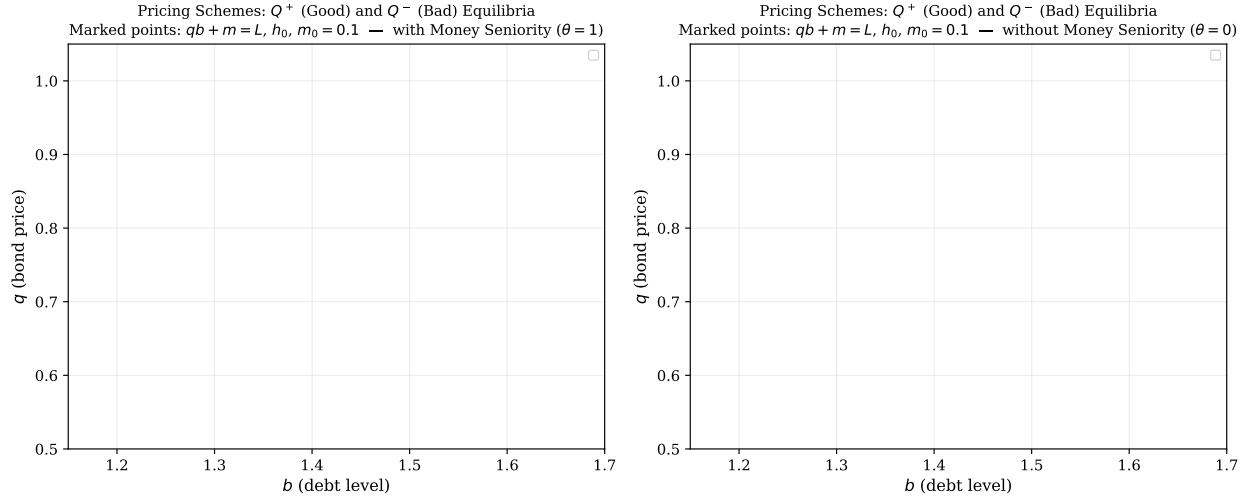


Figure 2: Unexpected QE Effect. *Left*: with reserves seniority ($\theta = 1$); *right*: without seniority ($\theta = 0$). Each panel shows Q^+ (solid) and Q^- (lighter) pricing curves for $m \in \{0, 0.10, 0.15\}$. Starting from $m = 0.10$, a buy-back that increases m and lowers b leaves the portfolio value $qb + m = L$ unchanged. With seniority, the pricing curves shift down significantly; without seniority, they barely move.

Proposition 3 (Benign Expansion Promises). *For any reserves seniority θ , a more aggressive balance sheet expansion $\Delta\beta > 0$ leads to two new equilibria $\tilde{Q}^+ > \tilde{Q}^-$ satisfying $\tilde{Q}^+ > Q^+$ and $\tilde{Q}^- > Q^-$. The new crisis region $[\tilde{b}^-, \tilde{b}^+]$ satisfies $\tilde{b}^- > b^-$ and $\tilde{b}^+ > b^+$. Reserves expansion slows debt growth and reduces the default premium despite increasing senior liabilities outstanding.*

The mechanism does not depend on reserves seniority, because the commitment operates through the flow budget constraint rather than through the recovery value Ψ . A slower debt accumulation path means that any given initial debt level is perceived as less dangerous: the probability of reaching the debt explosion region is lower, so investors accept a lower yield, validating the favorable equilibrium.

Figure 3 compares pricing curves for $\beta = 0$ and $\beta = 0.2$ under both seniority regimes. The shaded regions indicate the crisis zone for each β . Both Q^+ (upper curve) and Q^- (lower curve) shift upward with higher β , and the crisis region expands rightward — the economy can tolerate higher debt without falling into the bad equilibrium. The beneficial effect is qualitatively identical across the two seniority regimes.

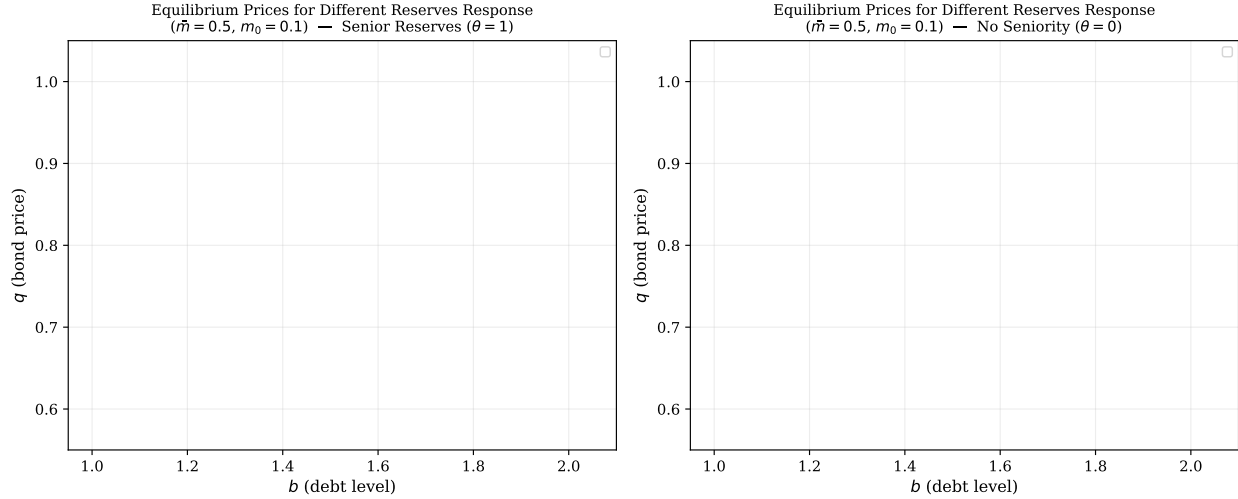


Figure 3: Pricing Functions with Different Reserves Response. $\beta = 0$ (blue) corresponds to constant money supply; $\beta = 0.2$ (red) to active reserves expansion. Shaded regions indicate the crisis zone (Q^+ vs. Q^- overlap) for each β . *Left*: senior reserves ($\theta = 1$); *right*: no seniority ($\theta = 0$). In both panels, higher β raises both equilibrium curves and enlarges the crisis-free region.

4 Conclusion

I study the sovereign default risk channel of quantitative easing in a continuous-time model of self-fulfilling debt crises. Reserves differ from short-term government bonds in that they are non-defaultable and senior to long-term bonds in the debt-resolution stage. These two features have opposite implications for balance sheet policy.

A one-time unexpected large-scale bond purchase financed by senior reserves *dilutes* existing long-term bond investors by reducing the net fiscal capacity available to repay bonds. The effect can be massively negative if the reserves injection is large enough to push the economy from the good to the bad equilibrium.

In contrast, a credible commitment to expand reserves at a faster pace — QE as a promise rather than a one-time action — slows the expected growth rate of long-term debt. This reduces the probability of reaching the crisis region, lowers the default premium, and raises bond prices, regardless of whether reserves carry seniority. The crisis region expands rightward: the economy can accommodate a larger debt stock without entering the self-fulfilling default zone.

These results isolate a real fiscal channel of QE that operates through the debt trajectory

and is distinct from the standard monetary-transmission channel (provision of short-term liquidity, manipulation of inelastic demand in fixed-income markets, or asset-price insurance). The model abstracts from inflation and from the general-equilibrium real-economy effects of QE, which is a natural direction for future research. Incorporating active monetary policy, heterogeneous investors, and endogenous fiscal capacity would enrich the model and allow it to speak to a broader set of QE programs.

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Appendices

Appendix: Asymptotic Analysis of the ODE System

To study asymptotic behavior I rephrase the system in terms of (q, v) where $v = qb$ is the market value of outstanding debt:

$$\begin{cases} (r + \delta + \lambda) q = \kappa + \lambda \Psi(b, \bar{m}) + \dot{q} \\ (r + \lambda) v = z - \bar{m} r + \lambda b \Psi(b, \bar{m}) + \dot{v} \end{cases} \quad (17)$$

where $\bar{m}$ is the (constant, post-cap) reserves stock.

Large- b asymptotics. When b is sufficiently high, the default probability approaches one and $\Psi(b, m) \approx (\phi/b) \mathbb{E}[(Z - \theta m)_+]$. Substituting, the system becomes

$$\begin{cases} (r + \delta + \lambda) q = \kappa + \lambda \frac{q}{v} \phi \mathbb{E}[(Z - \theta m)_+] + \dot{q} \\ (r + \lambda) v = z - \bar{m} r + \lambda \phi \mathbb{E}[(Z - \theta m)_+] + \dot{v} \end{cases} \quad (18)$$

Terminal conditions. The no-bubble terminal conditions for the bad equilibrium path are

$$q(T) = 0, \quad v(T) = \phi \frac{\bar{z} + \lambda \mathbb{E}[Z]}{r + \lambda}, \quad (19)$$

where T is the first time bond prices reach zero. The terminal value of v equals the discounted expected recovery value when the government enters permanent default with debt face value collapsing to the net present value of fiscal surpluses net of reserves obligations.

Proof of Proposition 1 (sketch). The demand curve $q_D(b) = [\kappa + \lambda \Psi(b, m)]/(\rho + \delta + \lambda)$ is strictly decreasing in b because $\Psi_b < 0$: higher debt increases the probability of default and reduces the expected recovery. The supply curve $q_S(b)$ is the level of q at which $\dot{b} = 0$. Differentiating implicitly, $q'_S(b) = [\kappa - h'(b)]/(\delta b) + q_S/b$. Since $h'(b) = \alpha_1 \in (0, 1)$ for $b < \bar{b}$ and $h'(b) = 0$ for $b > \bar{b}$ (where $\bar{b}$ is the kink point of the fiscal rule), the supply curve is non-monotone. Under the baseline calibration, q_D and q_S intersect twice, giving two steady

states. The eigenvalue analysis at the stable (left) steady state confirms it is a saddle point (one stable and one unstable manifold), validating the shooting construction. $\square$

Proof of Proposition 2 (sketch). An unexpected QE with $\Delta m > 0$ leaves $v = qb + m$ constant (the portfolio budget constraint (4)). Computing $\partial q / \partial m|_{v=\bar{L}}$:

$$\left. \frac{\partial q}{\partial m} \right|_{v=\bar{L}} = \lambda \frac{\Psi_m - (\Psi_b/q)}{(\rho + \delta + \lambda) - \lambda \Psi_q}.$$

With $\theta = 1$, $\Psi_m < 0$ and $\Psi_b < 0$ both work in the same direction (reserves seniority tightens the default condition through both the direct Ψ_m channel and the indirect Ψ_b channel via $b = (L - m)/q$). The numerator is strictly negative, so q falls with m at fixed L . Beyond the threshold $\bar{\delta}$, the new q falls below the Q^- curve, implying no good equilibrium exists at $(m + \Delta m, b + \Delta b)$. $\square$

Proof of Proposition 3 (sketch). With $\beta > 0$, the supply curve becomes $q_{S,\beta}(b) = q_{S,0}(b)/(1 + \beta) < q_{S,0}(b)$. The demand curve is unchanged. The stable (left) intersection shifts left (lower b) and the unstable (right) intersection shifts right (higher b): the crisis region expands. Since both intersections yield higher q (the demand curve is the binding constraint at intersections), both Q^+ and Q^- values increase. The mechanism is independent of θ because $q_{S,\beta}$ depends only on the flow budget constraint, not on the recovery function Ψ . $\square$